

In-Class Quiz: April 30

Points: 10/10

✓ **Correct** 1/1 Points

1. Consider a 5-stage in-order RISC pipeline consisting of Instruction Fetch (IF), Operand Fetch (OF), Execute (EX), Memory Access (MA), and Register Write (RW) stages. The pipeline does NOT implement data forwarding (bypassing). An instruction writes its result to the register file during the RW stage. A dependent instruction cannot enter the OF stage until the clock cycle *after* the producer instruction completes its RW stage. For the following code sequence, how many stall cycles (or NOP instructions) must the compiler insert between the two instructions to perfectly resolve the Read-After-Write (RAW) hazard?

I1: add r1, r2, r3

I2: sub r4, r1, r5 

☐ 1

☐ 2

☒ 3

☐ 4

✓ **Correct** 1/1 Points

2. A program containing a loop executes a conditional branch instruction (beq r1, r2, target). The processor uses a standard 5-stage pipeline (IF, OF, EX, MA, RW). Branch conditions and target addresses are evaluated and resolved entirely within the Execute (EX) stage. If the branch instruction is fetched in clock cycle 1 and evaluates to "taken", in which exact clock cycle will the correct target instruction be fetched, assuming the processor does not use branch prediction and simply fetches sequentially until the branch is resolved? 

☐ Cycle 2

☐ Cycle 3

☒ Cycle 4

☐ Cycle 5

✓ **Correct** 1/1 Points

3. A compiler is optimizing the instruction sequence below for a 5-stage pipeline (IF, OF, EX, MA, RW) without data forwarding. To prevent data hazards, dependent instructions must be sufficiently separated from their producers. If the compiler is allowed to reorder independent instructions without changing program semantics to minimize wasted cycles, what is the absolute minimum number of NOP instructions it must still insert before I4?

I1: add r1, r2, r3

I2: add r4, r5, r6

I3: add r7, r8, r9

I4: sub r10, r1, r4



☐ 3

☒ 2

☐ 1

☐ 0

✓ **Correct** 1/1 Points

4. Consider a processor executing the following code snippet:

I1: add r1, r2, r3

I2: sub r1, r4, r5

If the processor utilizes a strictly in-order pipeline, it will only ever experience Read-After-Write (RAW) data hazards. However, if the processor architecture is upgraded to allow out-of-order execution, what specific type of data hazard could arise between I1 and I2, and what is its fundamental definition? 

☒ A Write-After-Write (WAW) hazard, occurring if I2 writes to r1 before I1 writes to it.

☐ A Write-After-Read (WAR) hazard, occurring if I2 writes to r1 before I1 reads it.

- ☐ A Read-After-Write (RAW) hazard, occurring if I2 reads r1 before I1 writes to it.
- ☐ A structural hazard, occurring because both instructions attempt to access the ALU simultaneously.

✓ **Correct** 1/1 Points

5. In a standard 5-stage pipeline (IF, OF, EX, MA, RW), an arithmetic instruction `add r1, r2, r3` is fetched in clock cycle 1. During which clock cycle is the correctly computed mathematical result of this addition actively generated inside the Arithmetic Logic Unit (ALU)? 

- ☐ Cycle 5
- ☐ Cycle 4
- ☒ Cycle 3
- ☐ Cycle 2

✓ **Correct** 1/1 Points

6. To mitigate the performance penalty of control hazards, a processor architecture implements a "delayed branch" strategy with two delay slots. Which of the following statements rigorously defines the execution semantics of the instructions placed in these delay slots by the compiler? 

- ☐ They are evaluated conditionally and are committed to memory/registers only if the branch is taken.
- ☐ They are automatically flushed from the pipeline if the branch target is reached.
- ☐ They must exclusively be NOP instructions to ensure safe pipeline state transitions.
- ☒ They are assumed to be valid program instructions and are executed unconditionally, regardless of whether the branch is taken.

✓ **Correct** 1/1 Points

7. As an instruction flows through a 5-stage pipeline (IF, OF, EX, MA, RW), it carries an "instruction packet" containing state information from stage to stage via pipeline latches. Which of the following components is NOT logically present in the data transferred from the

Execute (EX) stage into the Memory Access (MA) stage via the EX-MA latch? 

- ☐ The program counter (PC) of the instruction.
- ☐ The mathematical result computed by the ALU (aluResult).
- ☐ The propagated control signals required for subsequent stages.
- ☒ The data fetched from memory (ldResult).

✓ **Correct** 1/1 Points

8. In a 5-stage RISC architecture, the Control Unit is situated in the Operand Fetch (OF) stage. It decodes the instruction and generates all necessary control signals for execution, memory access, and write-back. For a standard arithmetic instruction, where do these control signals physically travel immediately upon leaving the OF stage? 

- ☒ Into the OF-EX pipeline latch to travel synchronously with the instruction into the Execute stage.
- ☐ Directly to the Register File to trigger a data write.
- ☐ To the Instruction Fetch (IF) stage to update the branch target buffer.
- ☐ Directly to the Memory Unit to pre-fetch potential data operands.

✓ **Correct** 1/1 Points

9. Assume a program executes continuously on a completely un-pipelined processor, taking exactly T seconds to complete one full instruction cycle. If this processor is redesigned into a perfectly balanced 5-stage pipeline where each stage performs an equal fraction of the work, and the pipeline is kept completely full with no hazards, what is the theoretical time interval between the completion of consecutive instructions? 

- ☐ T seconds
- ☐ $5T$ seconds
- ☐ $T/25$ seconds
- ☒ $T/5$ seconds

✓ **Correct** 1/1 Points

10. A structural hazard occurs when two instructions attempt to utilize the same hardware resource in the exact same clock cycle. In a standard 5-stage pipeline (IF, OF, EX, MA, RW) with isolated instruction and data memories, which of the following combinations of simultaneous operations is structurally safe and will NOT cause a hazard? 

- ☐ Two consecutive instructions attempting to access the ALU in the EX stage.
- ☐ An instruction fetching an operand from the Register File in the OF stage while an older instruction writes a result to the Register File in the RW stage.
- ☐ An instruction fetching from instruction memory in the IF stage while an older instruction loads data from data memory in the MA stage.
- ☒ Both B and C are structurally safe in a standard 5-stage pipeline with dual-ported register files and separate memories.

